

Akriti Dhasmana

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Education

University of Colorado, Boulder	Boulder, CO
PhD in Computer Science and Engineering	2031
Advisors: Dr Maria Antoniak, Dr Alexis Palmer	
University of Notre Dame	South Bend, IN
MSc. in Computer Science and Engineering	2026
Coursework: Natural Language Processing, Large Language Models, Advanced Machine Learning.	GPA: 3.92/4.00
Union College	Schenectady, NY
BSc. in Computer Science and English, Minor in Mathematics	2022
Coursework: Data Science, Regression Analysis, Advanced Databases	GPA: 3.77/4.00
Awards: Summa Cum Laude, Dean's List, Presidential Scholarship, Scholar's Program Membership	

Research And Design Experience

University of Colorado, Boulder	Boulder, CO
Research Assistant	2026-2031
Assisting Dr Antoniak and Dr Palmer on ongoing projects related to narrative understanding and low-resource languages.	
University of Notre Dame	South Bend, IN
Research Thesis Student	2025-26
- Worked under the guidance of Dr. David Chiang on improving performance of speech recognition models such as XLS-R, HuBERT, wav2vec2 and wav2vec2BERT on low-resource languages and dialects for automated transcription tasks. - Developed methods for identifying suitable languages for cross-lingual transfer learning in automated speech recognition systems using phonological distances between similar languages and dialects.	
Chartbeat	New York, NY
Data Engineer II	2022– 25
- Implemented a recommendations engine geared towards helping publication platforms increase their readership. - Developed, maintained, and monitored the web-worker-queue architecture for up to a billion API requests at a time. - Designed and established a third-party integration with Google Search Console to fetch Search engine optimization metrics for Chartbeat customers. - Deployed applications in Python, C++, Java and Lua to be hosted on Kubernetes that interfaced with databases on RedShift, Postgres and Snowflake. Established metrics for measuring system performance and monitored backend services using Grafana and TSDB.	
Union College	Schenectady, NY
Research Thesis Student	2020-21
- Generated graphs mapping power structures between characters in self-written short fictional stories using Relational Character Analysis Tool (RCAT) and evaluated efficacy of common co-reference resolution tools such as NeuralCoref and AllenNLP on multilingual Hinglish texts. - Identified hierarchical social networks between fictional characters to surface social power dynamics reflected in text.	
Scholar Research Student	2020-21
- Designed and implemented a new data structure for cyclic abelian abstract groups with Dr. Matt Anderson. - Devised rules for mathematical operations such as addition, subtraction, and multiplication in mathematical groups. - Created the data structure in Python and evaluated performance across large order groups.	
Research Assistant	2019-20
Assisted Dr. Matt Anderson in his research to find larger Uniquely Solvable Puzzles to prove the existence of a faster method	

for matrix multiplication.

- Read and implemented a data structure based on Dr. S Minato's paper on Permutation Decision Diagrams to store and carry out mathematical operations on large permutations to generate Uniquely Solvable Puzzles.

International Business Machines

Yorktown Heights, NY

Summer Research Intern

2020

- Worked under the mentorship of Dr. Wendy Cornell, IBM Research Strategy Lead for Drug Discovery, to run a comprehensive literature analysis of drug discovery efforts related to COVID-19 using text mining and language processing techniques on aRxiv ChemRxiv, bioRxiv, PubChem and ChEMBL.
- I also attended regular meetings with the global community of interns and participated in knowledge enrichment sessions led by engineers at IBM.

Authored Publication

Dialect Matters: Cross-Lingual ASR Transfer for Low-Resource Indic Language Varieties

Rabat, Morocco

Authors: Akriti Dhasmana, Aarohi Srivastava, and David Chiang.

2026

In *Proceedings of the 13th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial)*, pp. 145–156. Association for Computational Linguistics.

Acknowledged Publication Contributions

Efficiently-Verifiable Strong Uniquely Solvable Puzzles and Matrix Multiplication

Hawaii, US

Authors: Anderson, Matthew, and Vu Le

2023

International Computing and Combinatorics Conference

Acknowledged Contributions: Ran experiments to find uniquely solvable puzzles

Matrix multiplication: verifying strong uniquely solvable puzzles

Alghero, Italy

Authors: Anderson, Matthew, Zongliang Ji, and Anthony Yang Xu.

2020

International Conference on Theory and Applications of Satisfiability Testing

Acknowledged Contributions: Ran experiments to find strong uniquely solvable puzzles

Conference Posters & Presentations

Dialect Matters: Cross-Lingual ASR Transfer for Low-Resource Indic Language Varieties

Urbana-Champaign IL

Midwest Speech Language Days, University of Illinois Urbana-Champaign

2026

The Billion-Request Content Recommendation System Challenge

New York City, NY

PyData Conference, Microsoft Conference Center

2023

Hierarchical Social Network Analysis of Short Fictional Texts

Schenectady, NY

Steinmetz Symposium, Union College

2022

Group Decision Diagrams (GDDs): A Data Structure for Mathematical Group

Schenectady, NY

Steinmetz Symposium, Union College

2021

Group Decision Diagrams (GDDs): An Efficient Data Structure for Groups

Virtual

ACM New York Conference for Women in Computing

2021

π DD: Permutation Decision Diagrams

Schenectady, NY

New York Six Research Consortium

2019

Awards and Fellowships

University of Colorado Boulder

Boulder, CO

- **Awtar and Teji Singh Graduate Fellowship:** Established in the memory of Awtar Singh Ph.D. '66, this fellowship is awarded to an Indian origin first-year student in Civil or Computer Science.
- **CEAS LSAMP Bridge Program:** Louis Stokes Alliances for Minority Participation Bridge to the Doctorate (LSAMP-BD) Program supports students with mentoring, networking and travel opportunities to increase the quantity and

quality of STEM graduate students from underrepresented populations.

Union College

Schenectady, NY

- **Phi Beta Kappa Membership:** National academic honor society recognizing exceptional achievement in the liberal arts and sciences, awarded to about 10% of graduates from the top 10% of U.S. colleges and universities.
- **Sigma Xi Membership:** Was nominated and inducted into Scientific Research Honor Society recognizing excellence in scientific investigation and research potential.
- **Alice P. and Donald C. Loughry Prize in Computer Science:** Awarded to one student for an outstanding senior project
- **Computer Science Department Service Prize:** For exemplary service and contributions to the CS Department.
- **Omicron Delta Kappa:** National leadership honor society recognizing achievement in service, and campus leadership.
- **Sigma Tau Delta:** International English Honor Society recognizing excellence in literary scholarship and leadership.
- **Most Impactful Leader in Media and Publication Prize:** Awarded for leadership as Co-Editor-in-Chief of Concordiensis.
- **Eugene I. Yudis Prize:** Best piece of prose fiction produced by a student.

Harvard University

Cambridge, MA

- **WECode Technology Leadership Award:** Honored for leadership and impact in technology by the nation's largest student-run women-in-STEM conference at Harvard University.

Columbia University and New York University

New York, NY

- **hackNY Fellowship:** Participated in a program started by faculty from NYU and Columbia University to foster innovation and community among top student technologists while interning in New York's tech and startup ecosystem.

Teaching Experience

University of Notre Dame

South Bend, IN

Teaching Assistant

2025-26

- Supported a Discrete Mathematics course and a Data Structures course by providing targeted tutoring, detailed feedback on proofs, and consistent academic guidance.
- Graded assignments and clarified complex concepts to strengthen students' understanding and overall performance.

WeSpeakNYC

New York, NY

English Language Class Facilitator

2022-24

- Led weekly, conversational English classes for diverse groups of adult immigrants.
- Fostered a supportive and interactive learning environment to build students' confidence and speaking skills.
- Facilitated practical discussions on topics essential for life in New York City, including civic engagement, workers' rights, and accessing city services.

Union College

Schenectady, NY

Teaching Assistant

2021-22

- Served as the teaching assistant for a course on Game Development.
- Guided student teams through the complete development lifecycle, from initial concept and prototyping to final implementation and playtesting.

Leadership Experience

Chartbeat Diversity, Equity and Inclusion Committee

New York, NY

Member

2022-24

- Led weekly planning meetings to coordinate social events and holiday celebrations that strengthened workplace community and belonging.
- Organized a mentorship initiative connecting employees with aspiring applicants from underrepresented backgrounds to support equitable access to opportunities.
- Founded and facilitated a weekly DEI-focused book club to encourage dialogue, reflection, and learning.

Association for Computer Machinery's Council on Women

New York, NY

- President 2019-22
- Organized events for students in the Computer Science department to engage with faculty and learn about opportunities to do research on campus.
 - Facilitated technical workshops designed to equip students with essential skills for future careers in STEM.
 - Coordinated department-funded trips to conferences such as NYCWiC and WECODE to promote professional development and foster interest in Computer Science.
- Concordiense** Schenectady, NY
Co-Editor-in-Chief 2020-21
- Reported on major national, international, and campus news for Union College's long-standing student newspaper.
 - Supervised a team of writers and editors while overseeing the paper's first transition to fully digital publishing.
 - Led weekly editorial meetings to assign stories, guide reporting, and plan each issue.
- Union College Robotics Crew** Schenectady, NY
Programming Team Leader 2021-22
- Led a six-member programming team in weekly sessions to design and implement a flood-fill algorithm for a micromouse navigating a 10x10 maze.
 - Competed in the IEEE Regional Micromouse Competition, earning Third Prize for the project's performance.

Mentorship Experience

- University of Colorado Boulder** Boulder, CO
Computer Science Directed Reading Program 2026
- Mentor an undergraduate STEM student navigating their coursework in computer science.
 - Assign weekly readings related to the field of their interest and hold weekly meetings to discuss the readings
- University of Notre Dame** South Bend, IN
AWIS Mentor 2025-26
- Mentor an undergraduate STEM student navigating their coursework in preparation for a future career.
 - Hold weekly catch-up meetings and assist mentee with any applications or guide through any career decisions.
- Union College** Schenectady, NY
Orientation Advisor 2022-24
- Led engaging orientation activities for first-year students, ensuring a smooth transition to college life and fostering a welcoming environment.
 - Fostered a supportive and interactive environment to help students build strong friendships with their peers.
- Union College** Schenectady, NY
International Students' Orientation Leader 2021-22
- Assisted international students in acclimating to U.S. academic and cultural life, providing guidance on navigating the challenges of studying abroad.
 - Organized events and activities that promote community-building, while offering mentorship to students dealing with homesickness and cultural adjustment.
- Union College** Schenectady, NY
Meaning Making Mentor 2021-22
- Coordinated events and exercises designed to help students reflect on and shape their personal narratives through meaning-making activities.
 - Facilitated workshops as part of a program funded by the Mensa Foundation, supporting students in their personal and intellectual growth.

Skills

- Technical:** Python, Java, C, C++, Arduino Uno, Google APIs, Google Dev, Snowflake, Djangofly, RabbitMQ, Redis, Apache

Kafka, Docker, building/maintaining CD/CI pipelines, Grafana Monitoring, AWS, Kubernetes, Adobe InDesign, PostgreSQL, Git Version Control and LaTeX, Data Munging, Software Design/Architecture, Web scraping, Machine Learning, Database Management, Reinforcement Learning, Query Generation, Data Mining, Text Analysis, Website Management, Social Network Analysis, Large Language Models.

Familiar with Jira Workflow, On-Call Engineering Duties.

- **Language:** Hindi, English, Intermediate Sanskrit.